NIST P-256 / P-384 Pollard rho on one RTX PRO 6000: the ECC2K-130 GPU ladder applied to prime fields

2026-10-13

What this is. A copy, for browsing, of the measurements behind Section 10 of the ECC2K-130 paper (`paper/ecc2k130/`). The source of truth is `gpu/nist/RESULTS.md` on `main`: round 1, and the round-2 update merged in aburan28/crypto pull request 1778. This report computes nothing new: every value below is copied from those records, and ratios are quotients of two recorded values.

Setup (as recorded). One NVIDIA RTX PRO 6000 Blackwell Server Edition (cc 12.0, 188 SMs), CUDA 13.3, `-arch=sm_120`, run on Modal through `gpu/nist/modal_app.py`. Walk: r-adding Pollard rho with the negation-map fold, distinguished points at probability 2^{-24}, W walks per thread sharing one Fermat inversion, CIOS Montgomery field with $n' = 1$. Each sweep cell is one run of 8,192 iterations. Modal runs: PTX build `ap-39mdjitYFxkMRHk6pFdd2P`, round-1 sweep `ap-o766pq9fqlurbzSYcqZoe6`, round-2 sweep `ap-0KNqNM8D1J4YQn5z67Ucc0` (timed out at Modal's 4 h limit during the last `maxreg=160` cells; every reported cell completed).

Correctness on the device. Distinguished-point counts matched $walks \times iterations/2^{24}$ in every cell (for example 11,969 observed against 12,085 expected on the first P-256 run; 12,154 against 12,096 in the best cell). The CPU suite cross-checks the compile-time-W stepper bit for bit against the generic stepper.

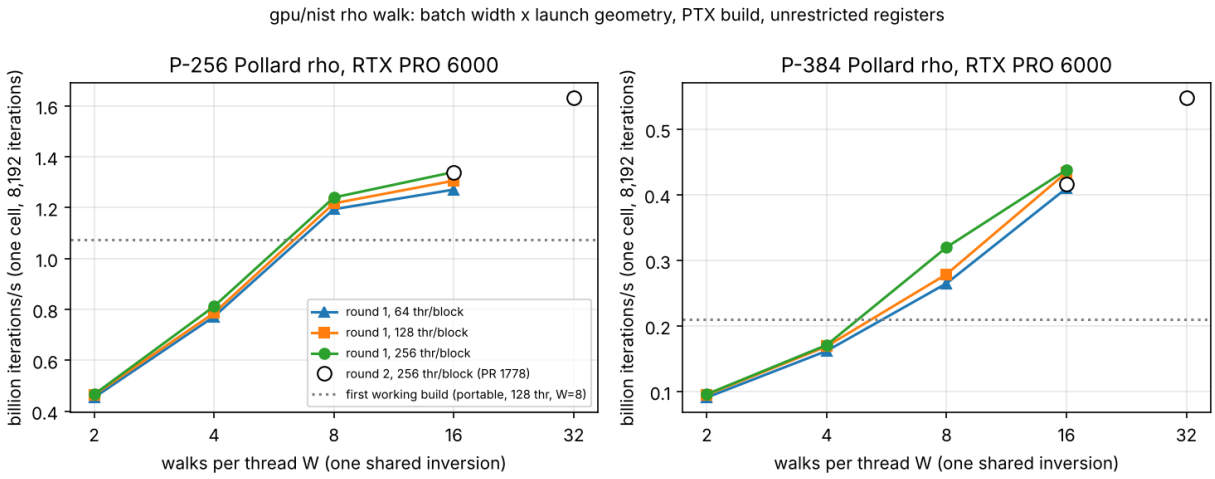


Figure 1: Recorded rho rates against the batch width W for three launch geometries (round 1) and the round-2 cells at 256 threads per block, unrestricted registers. Dotted line: first working build (portable, 128 threads, $W = 8$). Data: `data.csv`; plot: `plot.py`.

Limits. The cells are single runs, not interleaved repeated pairs, so no interval is available for any ratio. The round-2 gain cannot be split between the larger batch and the by-reference inversion because round 1 never compiled $W = 32$; at $W \leq 16$ the two rounds agree within noise. A Solinas-field walk was not built. At 1.63 billion iterations/s a P-256 collision needs about $1.9 \cdot 10^{29}$ s ($2^{127.8}$ iterations with the negation map): these are throughput measurements, not a statement about the security of the NIST curves.

step	cells compared	P-256	P-384
first working build	portable, 128 thr., $W = 8$	1.075	0.210
explicit PTX carry chains	same cell, PTX	1.282 (1.19×)	0.289 (1.38×)
$W = 8 \to 16$	256 thr., PTX	1.24 → 1.340	0.32 → 0.438
$W = 16 \to 32$, by-ref. inversion (PR 1778)	256 thr., PTX	1.339 → 1.632	0.417 → 0.548
register caps 96/128/160 (PR 1778)	256 thr., $W = 32$	1.589–1.595	0.447–0.455
best cell, cumulative	PTX, 256 thr., $W = 32$	**1.632 (1.52×)**	**0.548 (2.61×)**

Table 1: Rates in billions of iterations per second. Class: engineering; the step count is unchanged and $S = steps/\sqrt{n}$ is flat.